DEPARTMENT OF WORKS & HIGHWAYS

MATERIAL SCIENCE TECHNOLOGY BRANCH

Materials Testing Laboratory

[√] Port Moresby [ ] Lae, [ ] Mt Hagen [ ] Annex Lab.

Date:

Page of

SOIL PARTICLE DENSITY OF A SOIL - AS.1289.3.5.1

Client : ---------- Asset No.: ----------

Project: ---------- Material Source : ----------

Location: ---------- Pit No.: ----------

Material Description: ---------- Depth: ----------

Temperature of water _____ °C

Pw Density of water _____ g / cm3

Soil Particle Density of Fraction Passing 2.36mm Sieve

			C	D
M1	Mass of bottle + stopper	g		
M2	Mass of bottle + stopper + dry soil	g		
M3	Mass of bottle + stopper + contents	g		
M4	Mass of bottle + stopper + water	g		
Pf	Apparent Particle Densit $\frac{(M2 - M1) \times Pw}{(M4 - M1) - (M3 - M2)}$	gm / cm^3		
A	**Average Apparent Particle Density** (C+D) / 2	gm / cm^3		

Soil Particle Density of Fraction Retained 2.36mm Sieve

			E	F
M5	Mass of Oven Dried Sample	g		
M6	Mass of wire basket in water	g		
M7	Mass of wire basket and soil particle in water	g		
Pc	Apparent Particle Density (gm / cm^3) $\frac{M5 \times Pw}{M5 - (M7-M6)}$	gm / cm^3		
B	**Average Particle Density** (E+F) / 2	gm / cm^3		
	Soil Particle Density of Total Soil Samp (A+B) / 2	gm / cm^3		

Tested by: ---------- Checked by: ----------

Date: ---------- Date: ----------

Authorised by: AS (Material Science & Technology Branch)

FORM No.S-3.5.1

Revised Worksheet: 08/04/2022 lhb